Project Initialization and Planning Phase

Date	26/06/2026
Team ID	Not Assigned
Project Title	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The Rising Waters project proposes an intelligent flood prediction system powered by machine learning to address the growing threat of floods in India. By analysing historical meteorological data using advanced classification algorithms, the system delivers real-time flood risk assessments through an accessible Flask web application. Key features include an XGBoost-based prediction model achieving 96.55% accuracy, instant flood/no-flood classification, and cloud deployment for global accessibility.

Project Overview	
Objective	The primary objective of Rising Waters is to develop an intelligent, data-driven flood prediction system that leverages machine learning to accurately classify flood risk from meteorological inputs. The system aims to provide disaster management authorities, meteorologists, and government analysts with a reliable, real-time tool to support early warning decisions and reduce the devastating impact of flood events across India.
Scope	The project covers the complete machine learning lifecycle — from data collection and exploratory data analysis (EDA) to model training, evaluation, and web application deployment. It processes 10 meteorological features (temperature, humidity, cloud cover, annual and seasonal rainfall) to predict flood probability. The deployed system is accessible via a public URL on Render.com, making it usable by any authorized personnel from any location using a web browser.
Problem Statement	
Description	Floods are among the most devastating natural disasters in India, claiming thousands of lives and displacing millions every monsoon season. Current flood forecasting methods rely on manual meteorological analysis and threshold-based models that are too slow, inaccurate, and inaccessible for field-level disaster management officers. The absence of an intelligent, web-accessible flood prediction system leaves authorities underprepared and unable to issue timely warnings or deploy resources effectively.
Impact	Solving this problem will enable meteorologists to issue evacuation advisories several hours before a flood event, allow relief coordinators to prioritise resource deployment across multiple regions simultaneously, and provide government analysts with a validated, data-driven tool for operational flood preparedness. This will directly reduce loss of life, property damage, and economic losses caused by delayed or inaccurate flood warnings.
Proposed Solution	

Approach	The Rising Waters system employs supervised machine learning classification to predict flood risk from historical weather data. Four algorithms are trained and compared — Decision Tree, Random Forest, K-Nearest Neighbours, and XGBoost. XGBoost is selected as the final model due to its superior accuracy of 96.55% on test data. The trained model and StandardScaler are saved using Joblib (floods.save and transform.save) and integrated into a Flask web application that accepts real-time weather inputs and returns an instant flood/no-flood prediction with probability score.
Key Features	- XGBoost classification model achieving 96.55% accuracy on historical flood data. - Comparison of 4 ML algorithms with confusion matrix, classification report and ROC-AUC evaluation. - StandardScaler preprocessing saved separately (transform.save) for consistent real-time inference. - Flask web application with 4 HTML pages: home, prediction form, flood result, no-flood result. - Real-time flood risk classification with probability score and advisory action recommendations. - Cloud deployment on Render.com for global accessibility via public URL. - Version-controlled codebase on GitHub with automated deployment pipeline.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	Processor specifications	Intel Core i3 or above (i5/i7 recommended for faster training)
Memory	RAM specifications	Minimum 4 GB RAM (8 GB recommended)
Storage	Disk space for data, models, and notebooks	Minimum 2 GB free disk space
Network	Internet connectivity	Required for dataset download and Render.com cloud deployment
Software		
Frameworks	Web application framework	Flask 2.2+ with Gunicorn for production deployment
Libraries	Data science and ML libraries	NumPy, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn, Joblib
Development Environment	IDE and notebook environment	Anaconda Navigator, Jupyter Notebook, VS Code
Operating System	Supported platforms	Windows 10/11, Ubuntu Linux, macOS 10.15+
Data		
Dataset	Source, size, format	Kaggle — arbethi/rainfall-dataset flood dataset.xlsx, 115 records, 10 features, Excel format